PVSYST
PHOTOVOLTAIC
SOFTWARE

PVsyst TRIAL

PVsyst - Simulation report

Grid-Connected System

Project: rio-de-janeiro

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 4200 Wp

Rio de janeiro - Brazil

Author

PVsyst V8.1.7
VC0, Simulation date:
16/10/26 16:57
with V8.1.7

Project summary

Geographical Site
Rio de janeiro
Brazil

Situation

Latitude	-22.9110 °(S)
Longitude	-43.2094 °(W)
Altitude	2 m
Time zone	UTC-3

Project settings
Albedo 0.20

Weather data
Rio de janeiro
NASA POWER satellite data 2001-2020 - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

Orientation #1
Horizontal plane

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

System information

PV Array		Inverters	
Nb. of modules	7 units	Nb. of units	1 unit
Pnom total	4200 Wp	Total power	4.2 kWac
		Pnom ratio	1.00

Results summary

Simulation step	Hourly				
Produced Energy	6202.5 kWh/year	Specific production	1477 kWh/kWp/year	Perf. Ratio PR	85.94 %

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General parameters

Grid-Connected System — **No 3D scene defined, no shadings**

Simulation step
Hourly

Orientation #1
Horizontal plane

Models used

Transposition	Perez
Diffuse	Perez, Meteonorm
Circumsolar	separate

Horizon
Free Horizon

Near Shadings
no Shadings

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	Generic
Model	JKM-600N-72HL4-V
(Original PVsyst database)	
Unit Nom. Power	600 Wp
Number of PV modules	7 units
Nominal (STC)	4200 Wp
Modules	1 unit x 7 In series
At operating cond. (50°C)	
Pmpp	3896 Wp
U mpp	285 V
I mpp	14 A

Inverter

Manufacturer	Generic
Model	MIN 4200TL-X
(Original PVsyst database)	
Unit Nom. Power	4.20 kWac
Number of inverters	1 unit
Total power	4.2 kWac
Operating voltage	80-550 V
Pnom ratio (DC:AC)	1.00
Power sharing within this inverter	

Total PV power

Nominal (STC)	4.20 kWp
Total	7 modules
Module area	18.1 m²
Cell area	16.7 m²

Total inverter power

Total power	4.2 kWac
Number of inverters	1 unit
Pnom ratio	1.00

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 342 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss
Loss Fraction -0.75 %

Module mismatch losses
Loss Fraction 2.00 % at MPP

Strings Mismatch loss
Loss Fraction 0.15 %

IAM loss factor
Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

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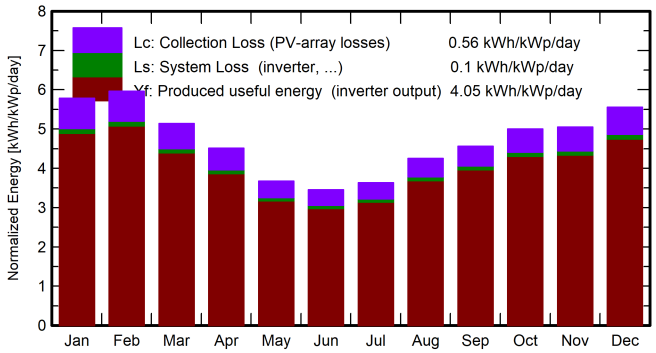
Main results

Simulation step
Hourly

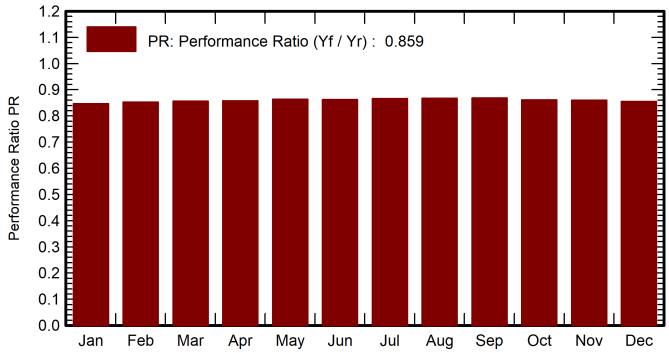
System Production

Produced Energy	6202.5 kWh/year	Specific production	1477 kWh/kWp/year
		Perf. Ratio PR	85.94 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	179.2	80.55	25.67	179.2	175.1	653.7	637.7	0.847
February	167.1	65.58	25.83	167.0	163.2	612.1	598.2	0.853
March	159.4	67.58	25.41	159.3	155.1	586.7	573.3	0.857
April	135.4	50.82	24.32	135.4	131.6	500.1	487.7	0.857
May	113.9	45.27	22.31	113.8	109.6	424.0	413.3	0.864
June	103.7	37.07	21.51	103.6	99.2	385.6	375.7	0.863
July	112.7	36.03	20.95	112.5	108.1	420.0	409.4	0.866
August	131.9	41.27	21.25	131.9	127.8	492.8	480.7	0.868
September	137.0	49.88	21.96	137.0	133.4	512.3	499.9	0.869
October	155.0	68.01	22.93	155.0	151.1	574.7	560.6	0.861
November	151.4	75.97	23.43	151.4	147.6	560.8	547.0	0.860
December	172.4	80.81	24.71	172.3	168.2	634.2	618.8	0.855
Year	1719.3	698.83	23.34	1718.4	1670.0	6357.0	6202.5	0.859

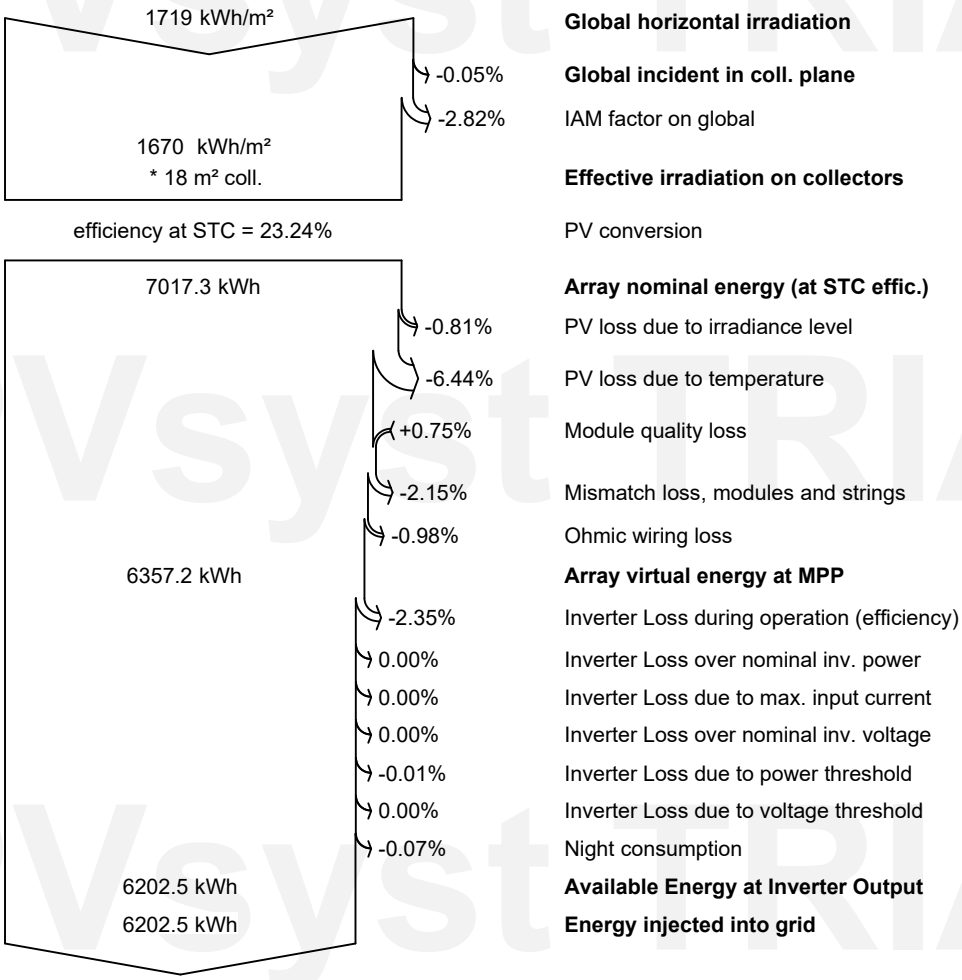
Legends

GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_Grid	Energy injected into grid
T_Amb	Ambient Temperature	PR	Performance Ratio
GlobInc	Global incident in coll. plane		
GlobEff	Effective Global, corr. for IAM and shadings		

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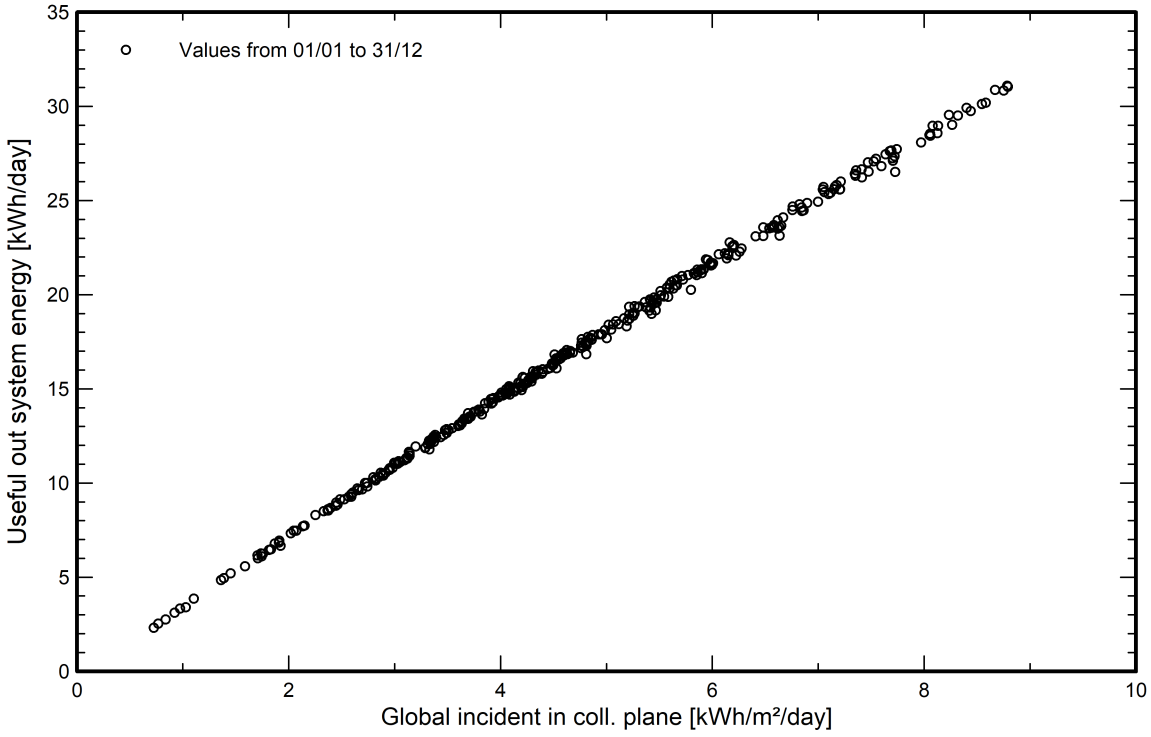
Loss diagram



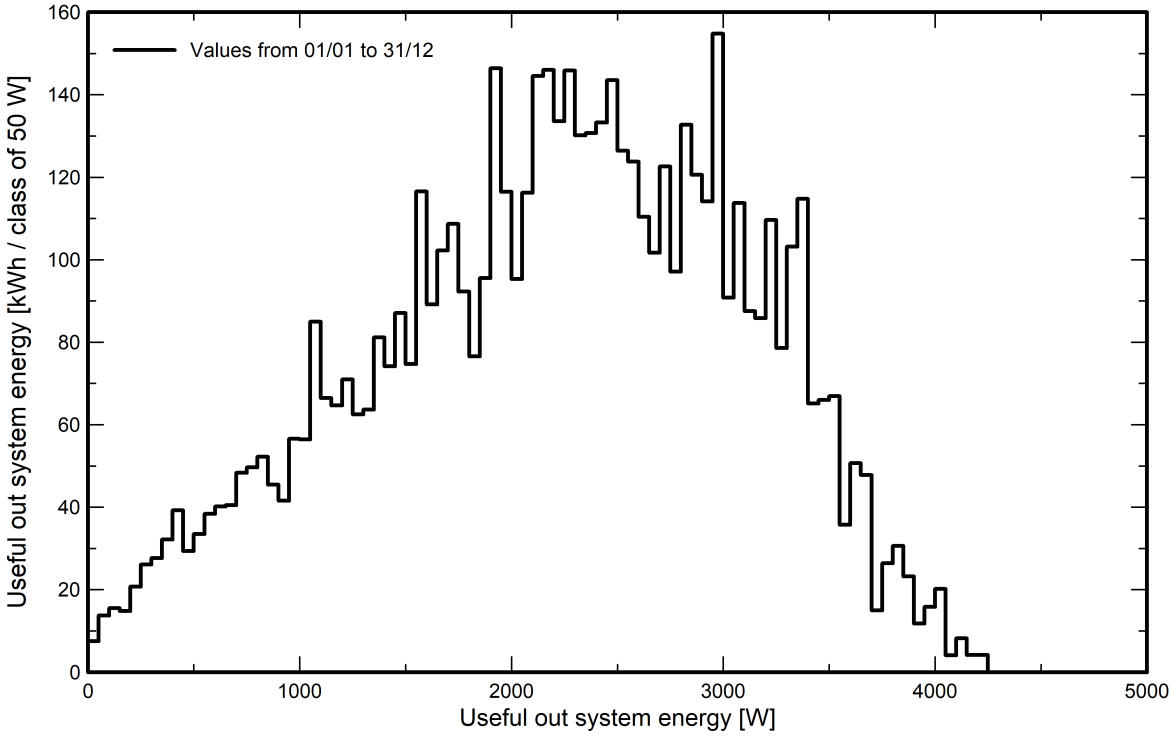
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Predef. graphs

Daily Input/Output diagram

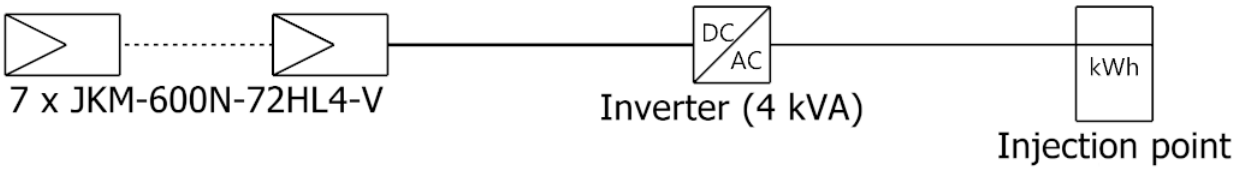


System Output Power Distribution



Single-line diagram

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PV module	JKM-600N-72HL4-V
Inverter	MIN 4200TL-X
String	7 x JKM-600N-72HL4-V

	rio-de-janeiro	
	VC0 : New simulation variant	16/10/26